Name: _____ Register No: _____

Experiment 2

Question: Upload and analyze the data set given in csv format and perform data preprocessing and visualization.

Code

```
import json
import re
from collections import Counter

import matplotlib.pyplot as plt
import numpy as np
import pandas as pd
import seaborn as sns

def exp_2(csv_path=None):
    if csv_path:
        df = pd.read_csv(csv_path)       # upload / load your file
    else:
        rng = np.random.default_rng(1)
        n = 200
        df = pd.DataFrame({
            "Age": rng.integers(18, 60, n).astype(float),
            "Salary": rng.normal(50000, 12000, n).round(0),
            "Dept": rng.choice(["IT", "HR", "Sales", "Finance"], n),
            "Experience": rng.integers(0, 30, n).astype(float),
        })
        df.loc[rng.choice(n, 10, replace=False), "Age"] = np.nan   # missing values
        df = pd.concat([df, df.iloc[:5]], ignore_index=True)       # duplicates
        df.to_csv("sample_data.csv", index=False)
        df = pd.read_csv("sample_data.csv")

    # ---- Analysis ----
    print("Shape:", df.shape)
    print(df.head(), "\n")
    df.info()
    print(df.describe(), "\n")
    print("Missing values:\n", df.isnull().sum(), "\n")
    print("Duplicates:", df.duplicated().sum())

    # ---- Preprocessing ----
    df = df.drop_duplicates()
    for col in df.select_dtypes(include="number"):
        df[col] = df[col].fillna(df[col].median())
    for col in df.select_dtypes(include="object"):
        df[col] = df[col].fillna(df[col].mode()[0])
    df = pd.get_dummies(df, columns=["Dept"], drop_first=True) \
        if "Dept" in df.columns else df
    print("\nAfter preprocessing:", df.shape, "| missing =", df.isnull().sum().sum())

    # ---- Visualization ----
    num = df.select_dtypes(include="number")
    fig, ax = plt.subplots(1, 3, figsize=(15, 4))
    df["Salary"].plot.hist(bins=20, ax=ax[0], title="Salary Distribution")
    sns.boxplot(y=df["Age"], ax=ax[1]).set_title("Age Boxplot")
    sns.heatmap(num.corr(), annot=True, cmap="coolwarm", ax=ax[2])
    ax[2].set_title("Correlation")
    plt.tight_layout()
    plt.show()


exp_2()
```

Output

```
Shape: (205, 4)
    Age   Salary     Dept  Experience
0  37.0  53994.0       HR         4.0
1  39.0  42185.0       HR        21.0
2  49.0  60349.0  Finance        20.0
3  57.0  48493.0    Sales         6.0
4  19.0  58030.0       HR        29.0

<class 'pandas.DataFrame'>
```

```
RangeIndex: 205 entries, 0 to 204
Data columns (total 4 columns):
 #   Column      Non-Null Count  Dtype
---  ------      --------------  -----
 0   Age         195 non-null    float64
 1   Salary      205 non-null    float64
 2   Dept        205 non-null    str
 3   Experience  205 non-null    float64
dtypes: float64(3), str(1)
memory usage: 6.5 KB
              Age        Salary  Experience
count  195.000000    205.000000  205.000000
mean    39.400000  48727.448780   14.648780
std     11.961881  11408.410361    8.534449
min     18.000000  17465.000000    0.000000
25%     29.000000  41242.000000    8.000000
50%     39.000000  48664.000000   15.000000
75%     50.000000  55832.000000   21.000000
max     59.000000  87201.000000   29.000000

Missing values:
 Age           10
Salary         0
Dept           0
Experience     0
dtype: int64

Duplicates: 5

After preprocessing: (200, 6) | missing = 0
```

Graph





